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REPORT

Software Design

Assignment 1

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Contents

1 Usecase Digrams **4**
1.1 Indicate usecases and actors 4
1.2 Usecases details 5
1.2.1 Make reservation 5
1.2.2 Pay for reservation 5
1.2.3 Cancel reservation 5
1.2.4 Pickup car 5
1.2.5 Manage incident 5
1.2.6 Report incident 5
1.3 Usecase diagram 6
1.4 Detail of use cases associated to the action "pick up car" 7
1.4.1 System sequence diagram 7
1.4.2 Lo-Fi interface 8

2 Class Diagram **14**

3 Sequence Diagram **15**
3.1 Reservation action 15

List of Figures

1.1 General usecase diagram 6
1.2 7
1.3 Main screen of Lo-fi inteface for pickup action 8
1.4 Reservation details for the pickup action 9
1.5 Select and allocate an available car in the booked category 10
1.6 Check the driving licence and store its scan 11
1.7 Record the joint inspection and the customer's digital signature 11
1.8 Record payment for a pay-at-pickup reservation 12
1.9 Review completed checks before handing over the keys 13

2.1 Class diagram 14

List of Tables

1.1 Actors in system 4
1.2 List of usecases and their participations 4
1.3 Make reservation usecase details 5

Chapter 1

Usecase Digrams

1.1 Indicate usecases and actors

From the description of the application, we defined the main actors of our application.

Actors	**Reason**
Customer	As mentioned in the assignment: "...customer starts a reservation by mentioning where they want to pick up the car..."
Agency Employee	As mentioned in the assingment: "Based on the chosen category, the agency employee will allocate a specific car to the customer."
Hotline Operator	As metioned in the assignment: "The 24/7 hotline will contact her back and organize with her the repairing actions"

Table 1.1: Actors in system

Design choices: Each represent a role interacting with the system. We choosed to design the system with one Customer for the actors before and after signing in because the action of authentication change the customer's access to the system, not changing their role. Agency Employee represent the employee working at the agency location and handle pickup and return action. Hotline operator handle report and coordinate assistances following an incident.

These actors with participate in several usecase:

Usecase	**Participation**
Make reservation	Customer
Pay for reservation	Customer
Cancel reservation	Customer
Pickup car	Customer, Agency Employee
Return car	Customer, Agency Employee
Manage incident	Customer, Hotline Operator
Report incident	Customer

Table 1.2: List of usecases and their participations

1.2 Usecases details

1.2.1 Make reservation

Name:	Make Reservation
Short description:	A customer reserve a car from the website of the company.
Precondition:	None
Postcondition:	A car booking is made; payment is completed for prepay.
Error situations:	None
System state in the event of an error:	No confirmed booking is created
Actors:	Customer
Trigger:	Customer wanting to rent a car.
Standard process:	(1) Customer goes to the website and starts a reservation. (2) Customer enters the pickup location and pickup and return date. (3) The system presents car category and their prepay and pay-later prices. (4) Customer select the car category. (5) Customer select the payment option. (6) Customer optionally choose soome extra insurances and options. (7) Customer signs in to continue the booking process. (8) Customer provides provides their email, first and last names, and address: street, city, country, and zip code. (9) If the customer has chose the prepay option, they then pay with credit card. (10) The system records and confirms the booking.

Table 1.3: Make reservation usecase details

1.2.2 Pay for reservation

1.2.3 Cancel reservation

1.2.4 Pickup car

1.2.5 Manage incident

1.2.6 Report incident

1.3 Usecase diagram

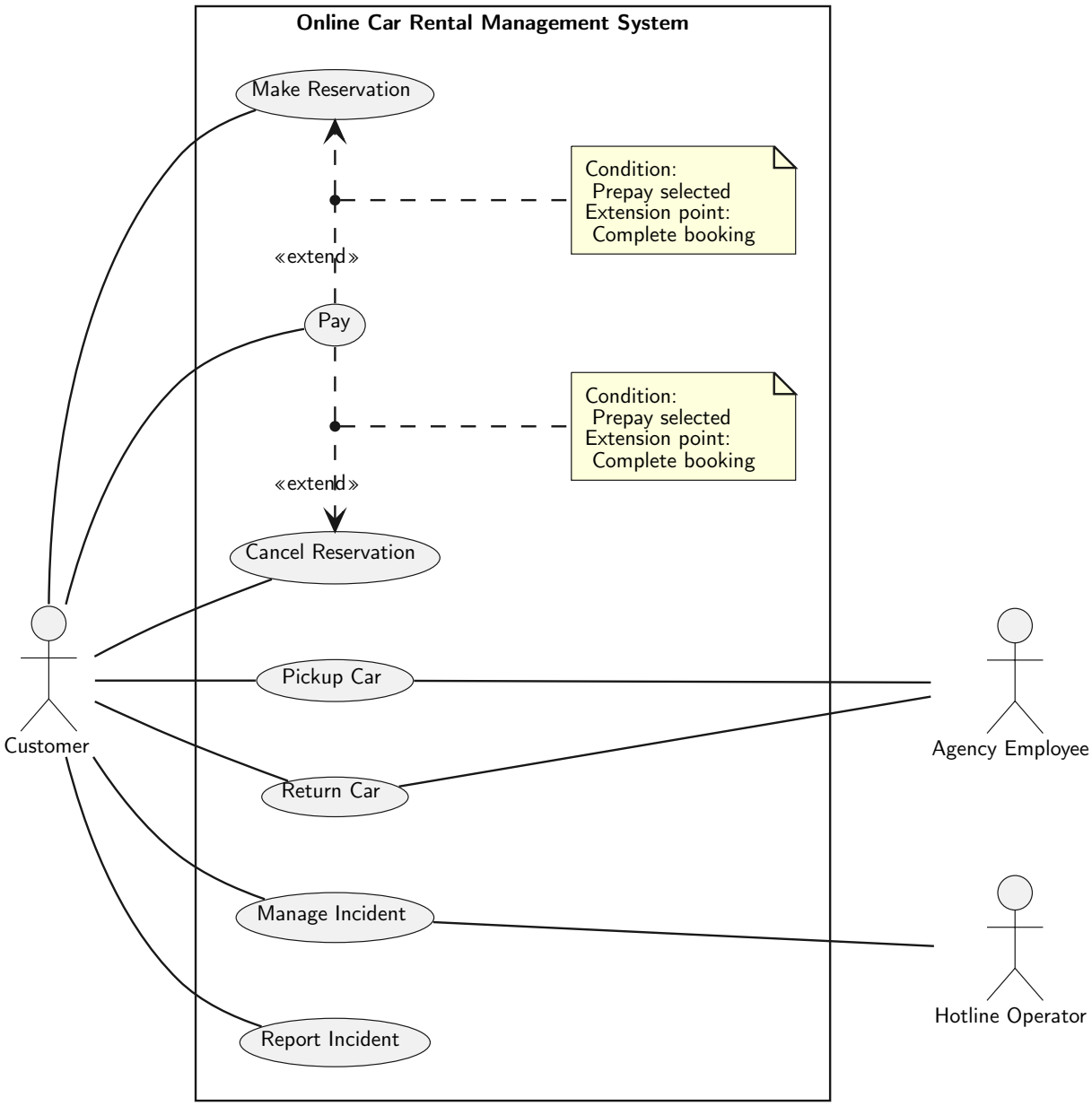


Figure 1.1: General usecase diagram

1.4 Detail of use cases associated to the action "pick up car"

1.4.1 System sequence diagram

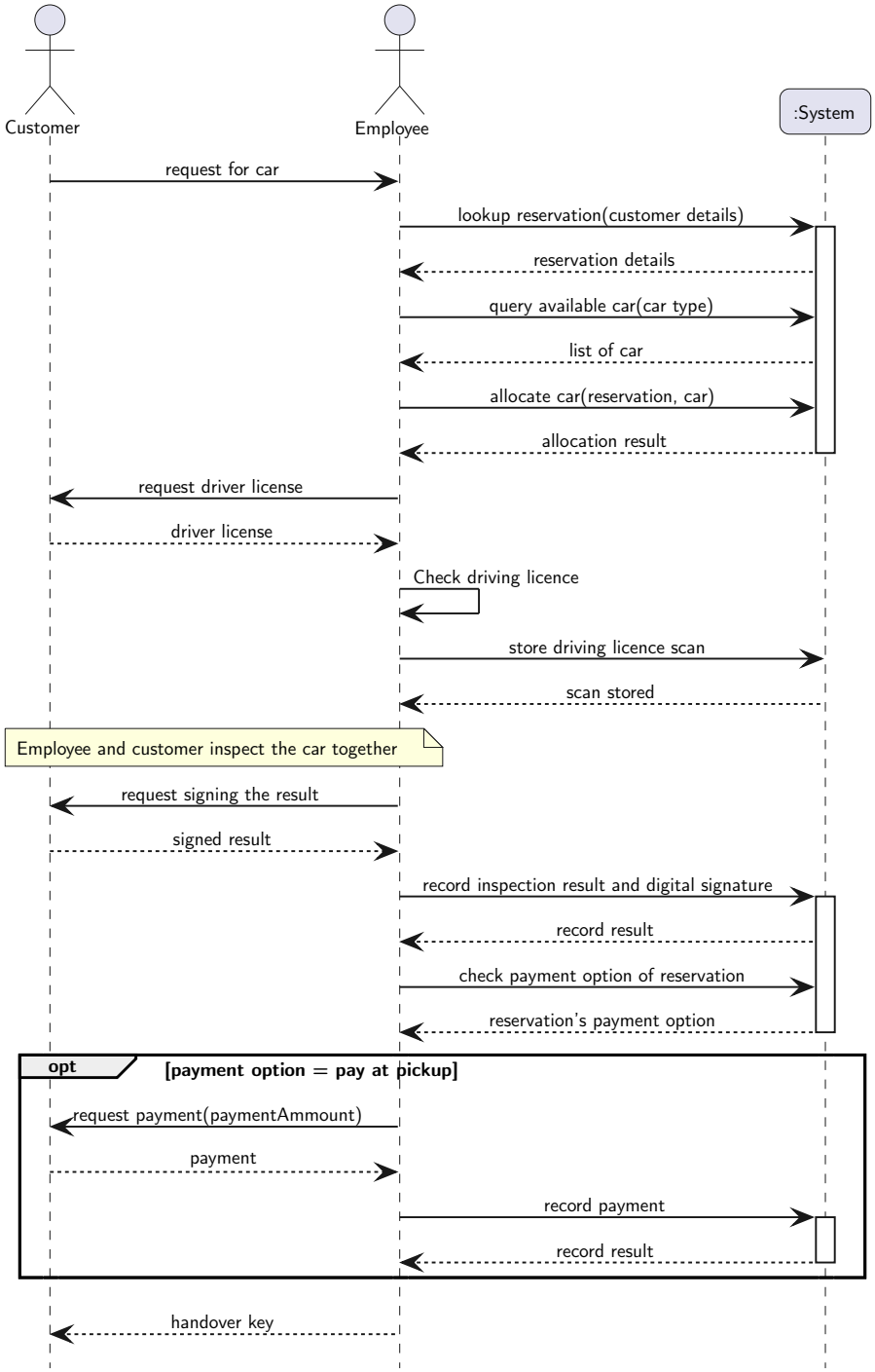


Figure 1.2:

1.4.2 Lo-Fi interface

The following screens follow the pickup sequence: find a reservation, review its details, allocate a car from the booked category, check and scan the driving licence, and record the joint inspection with the customer's digital signature. The examples use reservation RES-001 throughout. The payment screen applies only to pay-at-pickup reservations; after saving the inspection, a prepaid reservation with no balance due proceeds directly to the key-handover screen. Navigation buttons link to the corresponding figures in this PDF; they illustrate the workflow rather than execute it. Each save action must succeed before the employee continues, and the keys are handed over only after all checks and payment are complete.

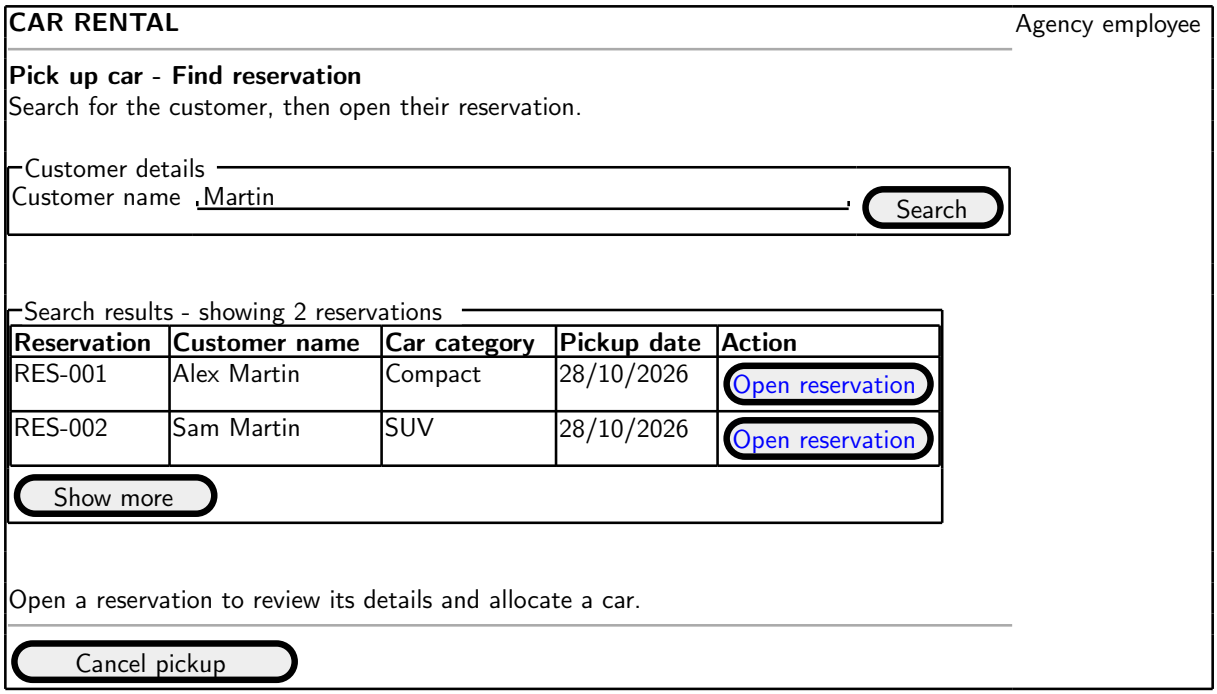


Figure 1.3: Main screen of Lo-fi inteface for pickup action

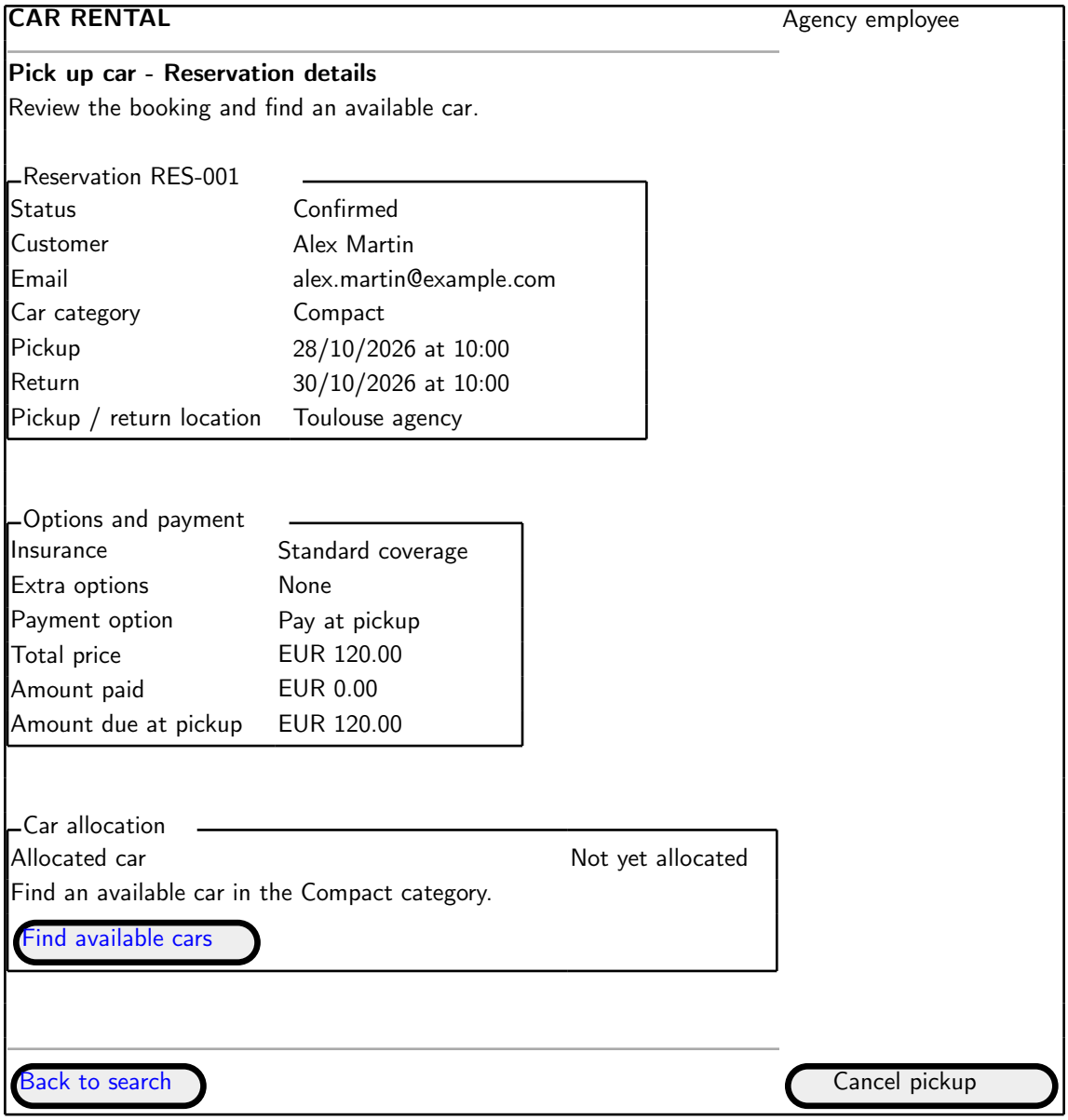


Figure 1.4: Reservation details for the pickup action

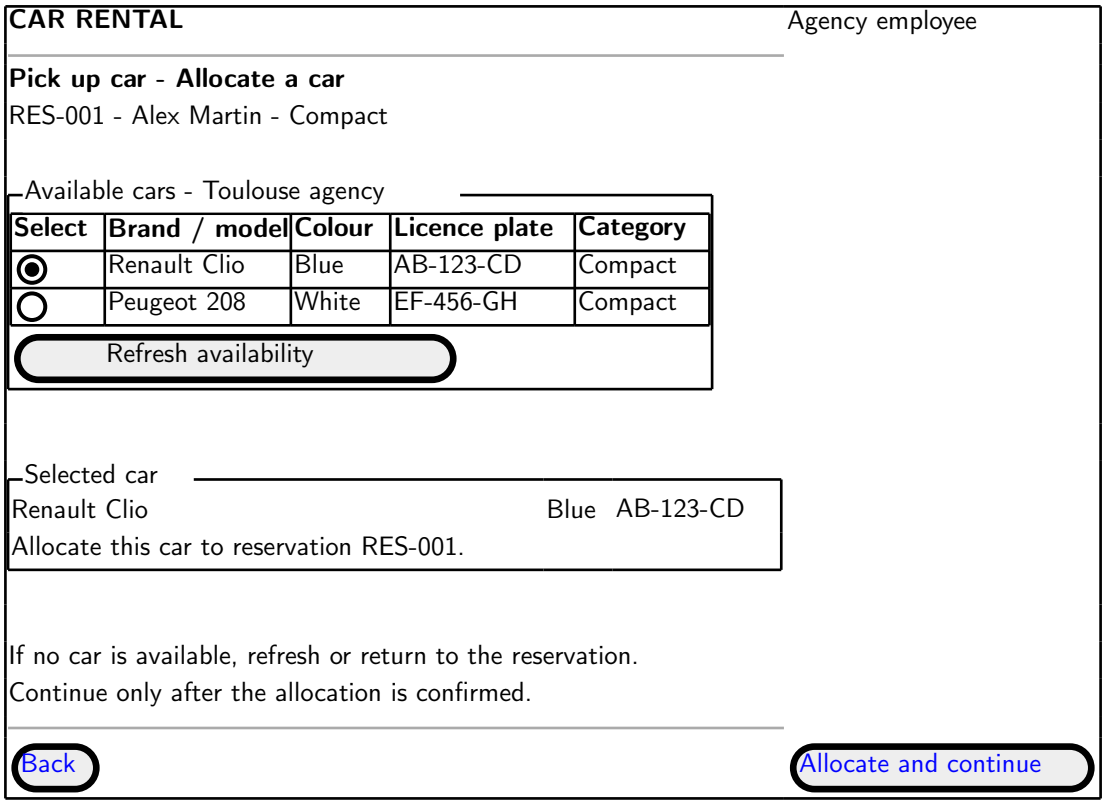


Figure 1.5: Select and allocate an available car in the booked category

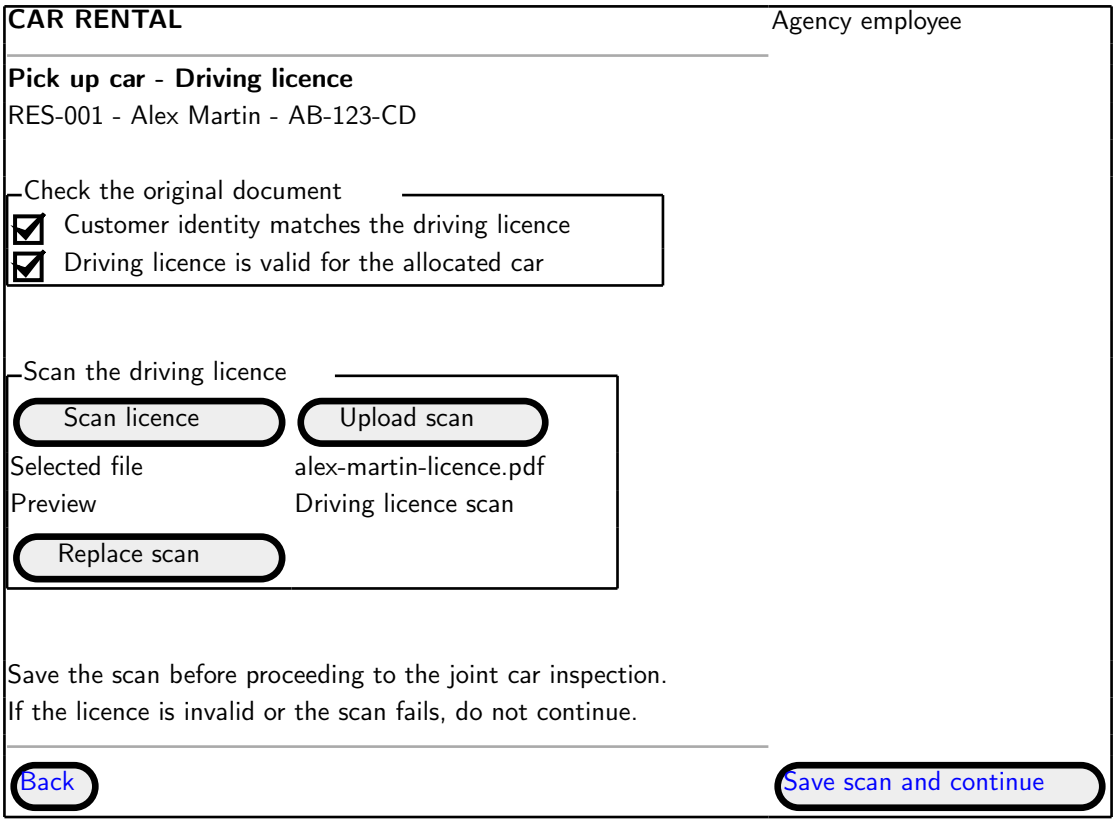


Figure 1.6: Check the driving licence and store its scan

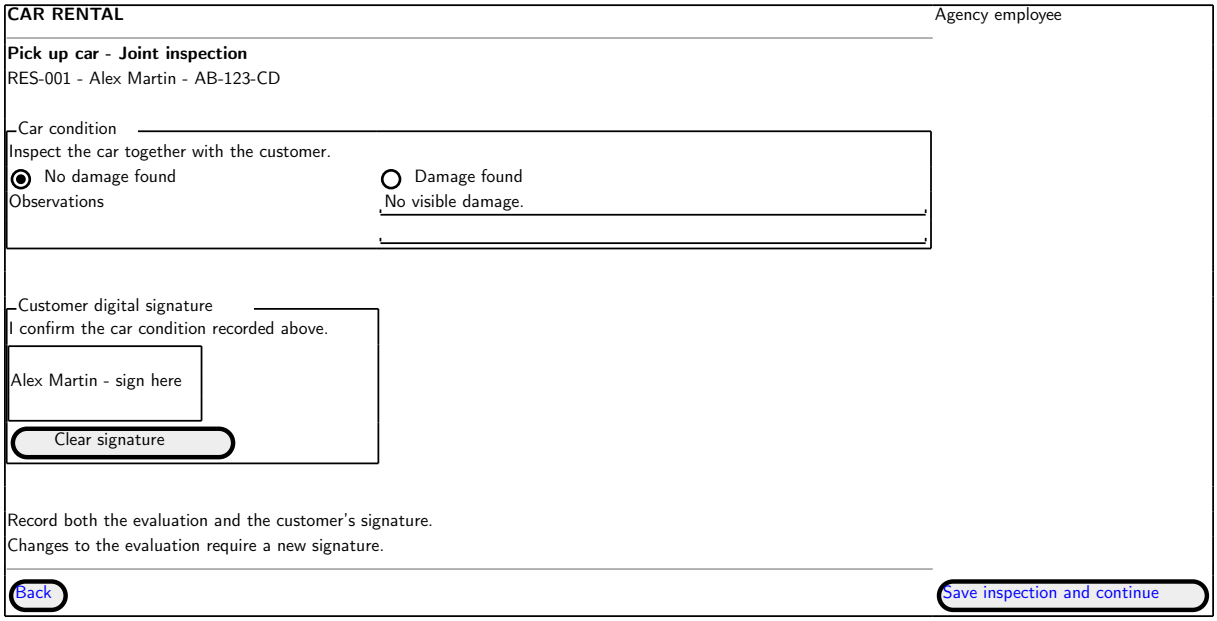


Figure 1.7: Record the joint inspection and the customer's digital signature

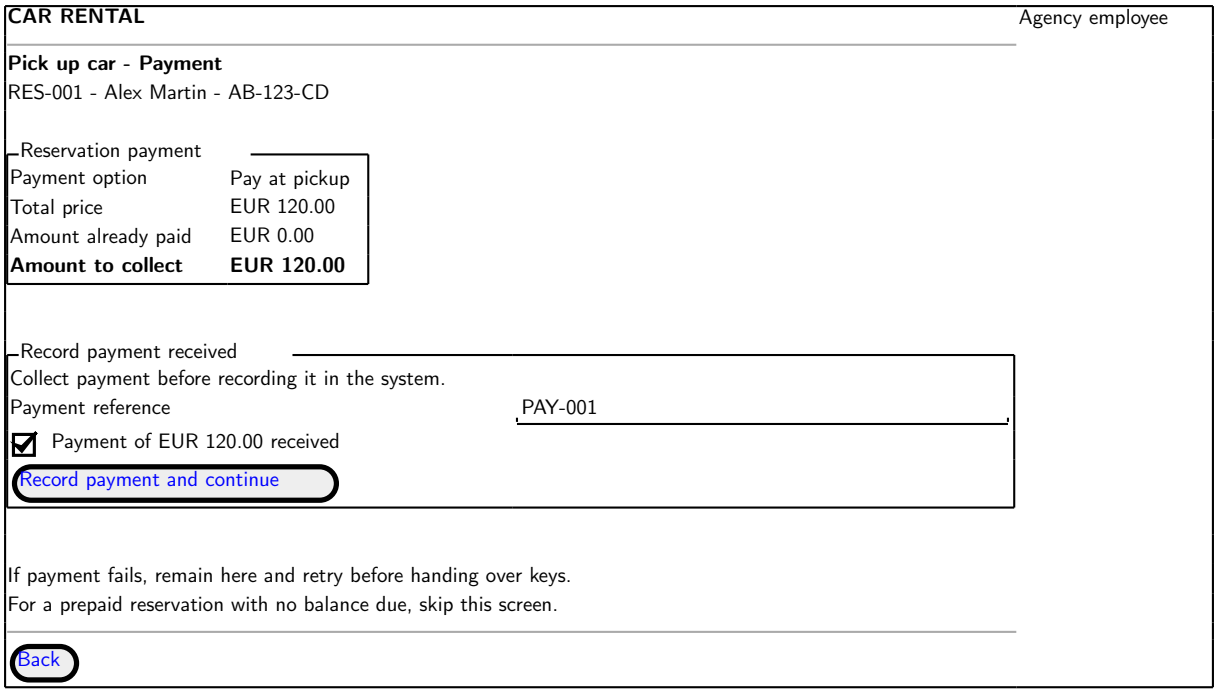


Figure 1.8: Record payment for a pay-at-pickup reservation

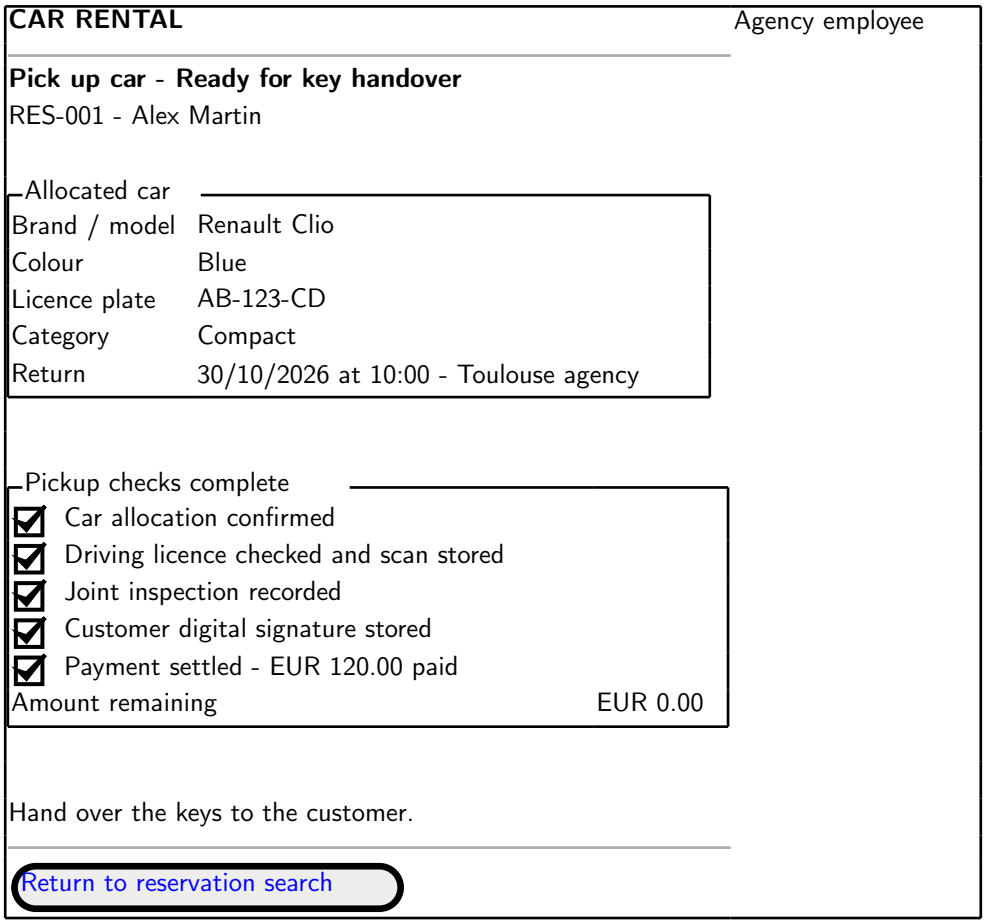


Figure 1.9: Review completed checks before handing over the keys

Chapter 2

Class Diagram

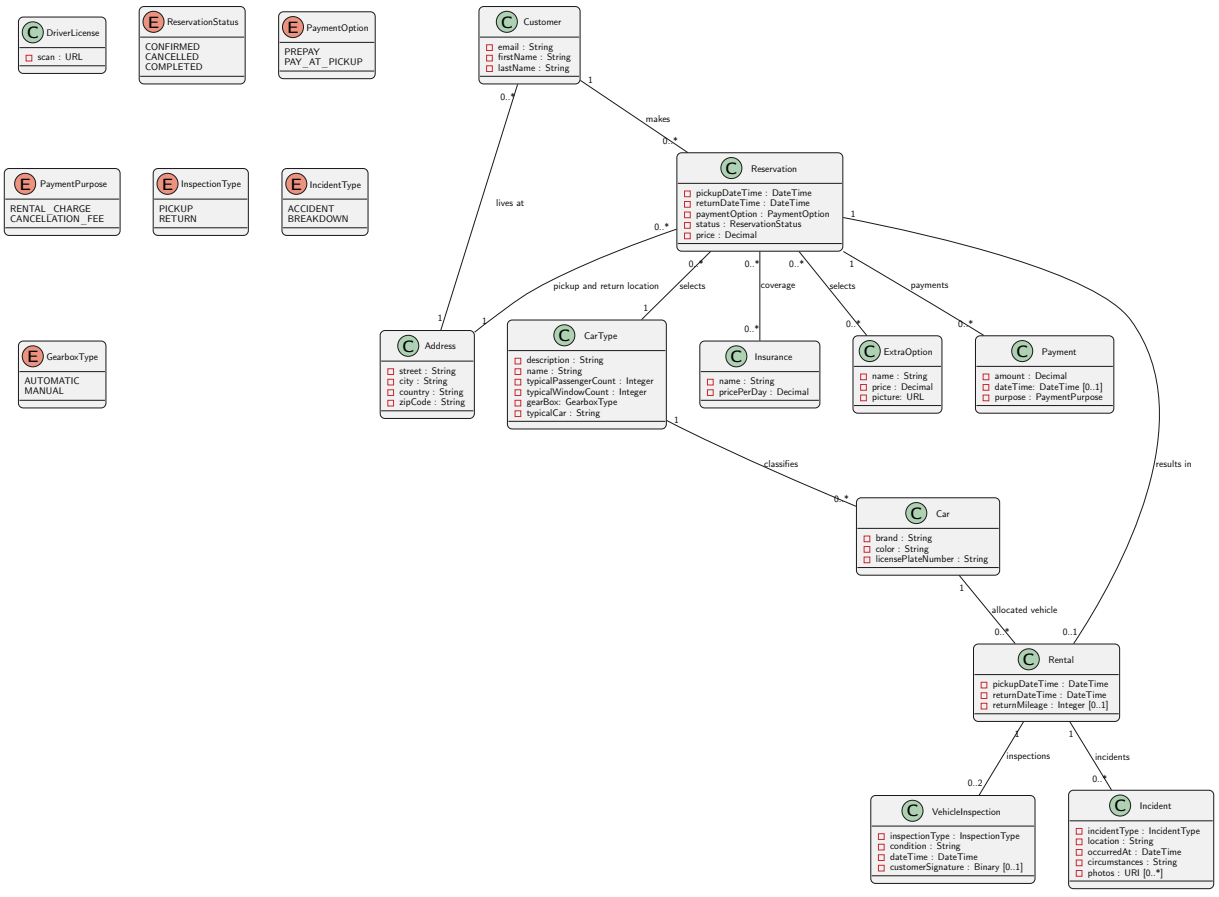


Figure 2.1: Class diagram

Chapter 3

Sequence Diagram

3.1 Reservation action

$$h_\theta(x) = \theta^T X \tag{3.1}$$

$$\text{ProbabilityOfClass}_1 = P(Y = 1|X) = \sigma(\theta^T X) = \frac{e^{\theta^T X}}{1 + e^{\theta^T X}} \tag{3.2}$$